

Robocup - Conceptual Design Report

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Due 16/05/2025

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1 Introduction

The aim of this report is to present the research and development of three feasible robot design options so that a final "best" design can be selected to move forward with. Through this report, the use of mechatronic system design techniques taught during the first semester of ENMT301 will be demonstrated through the use of research, concept generation, evaluation and comparisons. The goal of the overall project, is to design and build a robot in groups to compete head-to-head, completing a variety of tasks including navigating an unknown arena, identifying and avoiding plastic weights and identifying and collecting metal weights. Along with this a "snitch" may be collected to achieve a large amount of points. To achieve the aim, three concepts will be developed to a level that allows robust analysis and comparison, with each concept presenting a unique angle to meet the requirements.

The general approach to complete the initial design process starts with researching past competitions, defining specific requirement specifications, develop proposed concepts, test concept features to obtain preliminary results, evaluate concepts, and determine one concept to be developed. The design approach will begin with competition research, to observe past effective/ineffective strategies to implement. This would include identifying the main priorities of the robot. From this, requirement specifications can be developed to identify all important functions/behaviors that the design should achieve. These steps were used as a foundation for each member to design an individual concept, which covers the main functionalities and hardware of the robot. This generated multiple approaches which can be assessed to generate a final design. During this phase, features and components were tested to define their specific behavior. All three concepts were evaluated and compared using a decision matrix. Each compared figure of merit was based on requirements previously outlined. To determine the figures of merit, requirements were either considered based on attributes of the system or parts or derived through simulation. Where requirements were qualitative, appropriate surrogate models were considered. Based upon these evaluations, the concept with the highest average score will be selected to develop further.

2 Research

2.1 Harvester Collection

The winners of the 2024 competition, group 1, featured a combine harvester style weight collection system. This proved effective, with 2.5 weights collected on average. To further this, during the 2020 competition, 2 out of 8 competitors which had the ability to catch the snitch did catch the snitch; both of these were open mouth designs. This design consisted of an open mouth, with a spinning horizontal drum. The drums horizontal beams used rubber bands, which would stretch to pull the weight into the open mouth. Notable features that make this design feasible are as follows:

- A motor with high torque and speed, so that the weight can be thrown into the open mouth of the robot without further assistance.
- Rubber bands as horizontal bars, as these provide sufficient friction to grip the weights, and can stretch to the weights' shape.
- A large ramp behind the harvester to raise the weight before collection.

2.2 Robots stuck in the wall

Throughout the Robocup competition, there have been a variety of cases of robots getting stuck on the poles, the walls and themselves. In the 2024 Robocup competition, in round 12, robot 09 jammed itself on the pole. In round 11, Robot 21 had spent a long time navigating around the pole, whilst in round 57, Robot 01 beached itself, with 26 being stuck in a tight corner. Across the competition, there are a variety of occasions where the robot's square-like shape gets it stuck in various areas across the course. The minimum width between walls is defined at 400mm, this requirement would lead to obstruction if the robot was developed to be greater than that. From this observation, some different geometries were looked into for the robot's shape, shown in Figure 1 below.

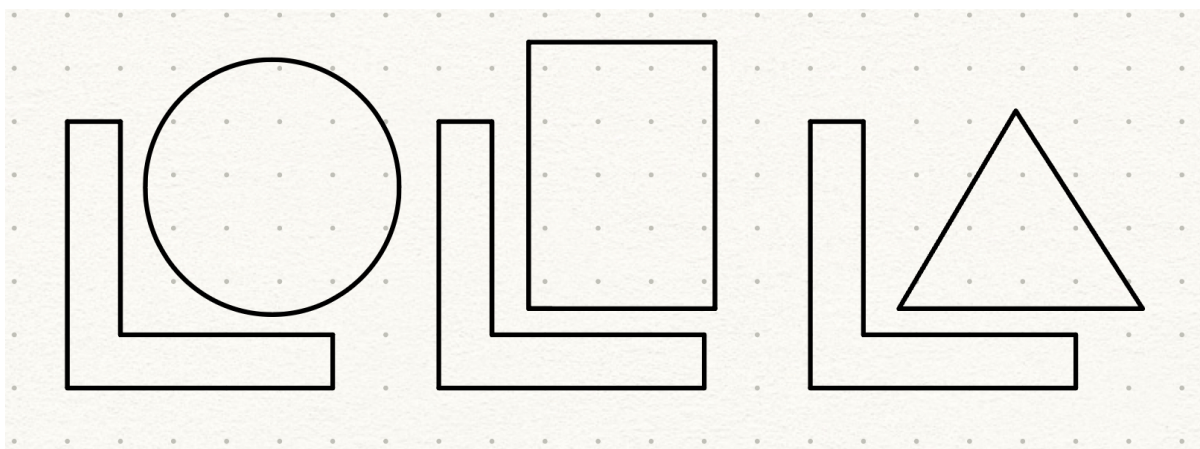


Figure 1: Shape testing layout as compared to the dimensional 400*400mm corner constraint to illustrate the circles ability to not get as easily stuck

From visual analysis of previous arena types, it was deemed to be required to take this one as an important consideration when designing the shape of the robot. Ensuring the robot would be able to traverse any portion of the arena became a significant detail.

2.3 Tracks vs Wheels

Viewing previous winning robots currently displayed in the main Mechatronics Lab showed that all utilised a track style movement system instead of any of the wheels provided. From visual observations from Robocup 2024 in rounds 4 and 36, robot 33 (an omniwheel design), struggled with maneuverability and limited traction, as compared to the tracked competition. Observations of the course observed small bumps and inclines. This kind of terrain would require the distributed traction that tracks provide. This pushed the designs further towards more track orientated approaches.

2.4 Inductive sensor inside of holding chamber

During the observation of previous winning robots, an approach our team was interested in was a discriminating collecting system within the robot, as opposed to during detection. This discriminating system would differentiate between plastic and metal weights. Instead of sensing the weights outside of the robot, the robot picked up all weights and tested inside the holding chamber, removing any plastic weights found through utilisation of an exit ramp. This was achieved using an inductive sensor, along with a gate-like system to dictate which location the weights would move to as shown in Figure 2 below.



Figure 2: Inductive sensor located inside of the 2024 Robocup winner, Robot 1

2.5 Internal sorting and weight orientation

When collecting weights, many various robots had geometries to help guide the weight into a certain orientation for sorting or storage. A great example would be the 2024 RoboCup winner. The horizontal ramp guides, paired with the initial velocity applied to the weight by the harvester, made the weight orient itself so that the inductive sensor would work effectively. Additionally, such orientation made it easier to sort into the weight storage basket.

2.6 Track tension and movement

From majority of previous winners and from video observation of the robots in the competition, it was found the tracks had to be stretched in weird and unique ways to maintain track tension, one of which is shown in Figure 3 below.

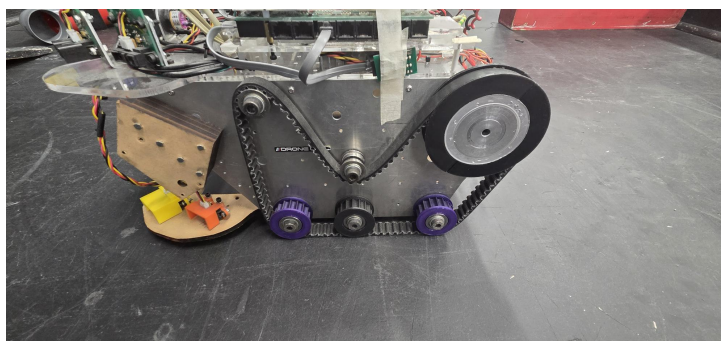


Figure 3: Past Robocup winner robot with track tensioning design

As from an investigation done onto the tracks, they are handed down between the years. Hence this means that the tracks are worn out by the previous years track configurations which can reduce movement performance. previous designs had to create more convoluted ways of enabling tension from the tracks, which made maintenance and design modification harder for those designs. This means a system not using tracks, or an efficient tensioning system would need to be considered. Along with this, the variations in track tension lead to a change in speed from each side of a test robot, causing rotation when not warranted. This meant that an efficient algorithm within code would need to be created to sustain forward alignment throughout the rounds.

3 Requirement Specifications

1. Performance Requirements

- 1.1 The robot should be able to pick up weights up to 1 kg (without need of initial discrimination)
Rationale: There is not enough available time in the match nor reliable enough design to try to discriminate the weight's mass and before pick up.
- 1.2 The robot should be able to detect a weight within 250mm of itself (without need of initial discrimination)
Rationale: The robot needs to have some guidance to weights to make it more effective at weight pickup.
- 1.3 The design must be able to determine what type of weight it has picked up in a maximum 5 second time frame
Rationale: If there are multiple weights that the robot encounters, the weight processing must be as efficient as possible.
- 1.4 The robot must be able to capture the snatch, even if that does sacrifice performance.
Rationale: The snatch is a valuable asset to capture during the competition, and its size makes it quite difficult to properly process without increased dimensional constraints.
- 1.5 The robot must be able to climb a 30 °incline and a maximum height of 100mm
Rationale: Previous competitors had large issues with ramps leading to large failures

2. Functional Requirements

- 2.1 The robot should be able to turn around in any corners without getting stuck
- 2.2 The robot shall be capable of turning 360 degrees while remaining in place
Rationale: These features allow the robot to be able to reorientate itself at any location in the arena
- 2.3 The robots' track system must be able to move the robot at walking pace (6.5km/h) when fully loaded.
Rationale: The time pressure of the match is a major factor, the faster the robot can traverse the course, the faster it can collect weights when considering competition.
- 2.4 Weights must be able to be picked up no matter orientation into the system
Rationale: During a round weights are knocked over and moved, this leads to a variety of orientations that would need to be picked up
- 2.5 The robot must keep track of the amount of weights on board the robot
Rationale: Holding more than 2 weights at the end of the round will lead to point deductions.

3. Operational Requirements

- 3.1 All components of the robot, mechanical or electrical, must be able to function for 10 minutes or more.
Rationale: The match only lasts a fraction of the time, however stress testing the robot with longer times and introducing stressful situations on the system may present deficiencies that can be fixed before the proper competition.

4. Non-functional Requirements

- 4.1 The robot's components must be easily removed and replaced to allow for modularity
Rationale: The time between rounds is short, if modification needs to be made it needs to be efficient and quick

5. Constraints

- 5.1 The robot must fit in a 400x400mm area.
Rationale: The tightest corners of the design are within that 400mm size, so a robot that can traverse in tight areas like that will be able to go across majority of the course fine.
- 5.2 All parts fabricated should be able to be fabricated at university with no external help or specific constraints.
Rationale: If failures occur during the competition, parts would need to be fabricated within the university.

4 Design Space and Concept Generation

4.1 Concept 1: Timezone, Charlie Horne

4.1.1 General brief

This design was inspired by the collection mechanism of claw machines. It uses a similar claw design to grab weights. A lever arm drops weights into a collection shoot in a similar way to claw machines. Weights are then checked as metal or plastic using an inductive proximity sensor and either deposited into the storage system or out of the robot using trapdoors. Once returned to base, the robot drops weights out the back.

4.1.2 Collection System

This concepts proposed collection system uses a claw design and a lever arm to collect weights. The claw will use two arms to close around and squeeze weights in any orientation. This will then be lifted by the lever arm into a collection shoot, orientating the weight as it falls into the sorting system. The claw will be held open when not in use by using a spring between the claw arms and a slider. To close the claw arms around a weight, a motor at the end of the lever arm will reel in a string attached to the slider using a pulley system. This will pull the slider towards the claws, causing them to close. This system has been developed so that the claw can be actuated using a motor with high torque and speed, while avoiding excess weight at the tip of the lever arm. The collection shoot will consist of an open mouth funnel, which will direct the weight as it falls from the claw into the storage system. It will be shaped so that it orientates the weight into the correct location without getting stuck. The claw arms will be made of MDF. The controls arms, slider and claw connection will be made of PLA. This assembly is seen in Figure 4 and Figure 5.

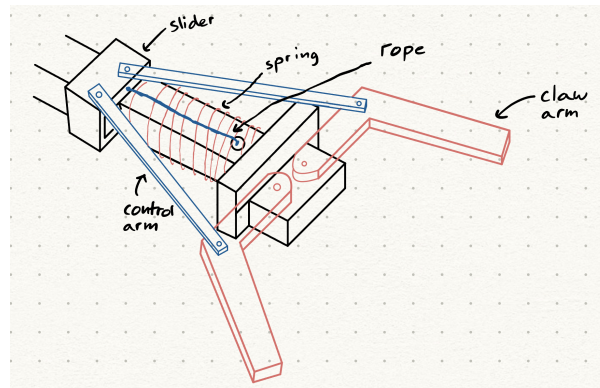


Figure 4: Claw concept for weight collection

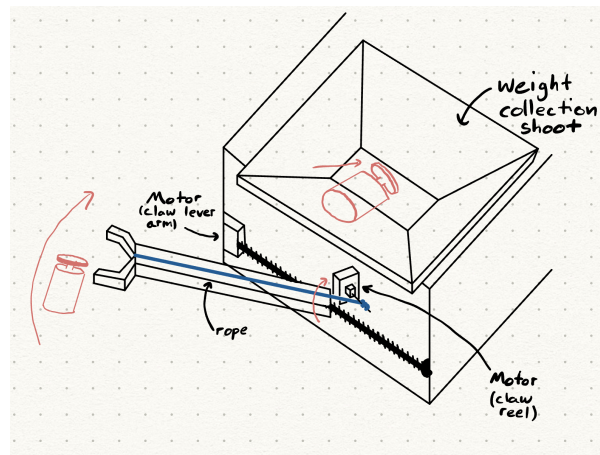


Figure 5: Collection shoot leading to sorting system

4.1.3 Sorting System

Weight will enter into the sorting system from above from a known orientation. The weights will then roll along a slanted floor towards a corner. The walls of this corner will consist of two trap doors. These walls will open to direct the weight depending on if the weight is detected to be metal or plastic. The side wall will lead outside the robot to dispose of plastic weights. Metal weights will fall through the front trapdoor. An inductive proximity sensor will be placed flush with the floor, and will check the material of the weight, as seen in Figure 6. The trapdoors and floor will be made of MDF.

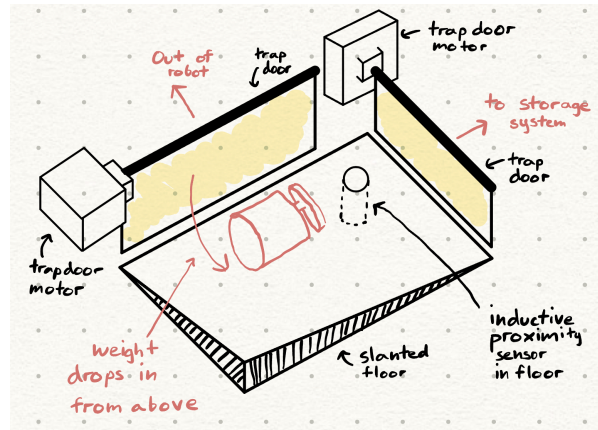


Figure 6: Trapdoor sorting system

4.1.4 Storage System

The storage system will consist of an empty space room with an opening from the storage system on one wall, and a trap door on the other. The floor will be slanted so that weights roll against the trap door. Once returned to base, the trap door shall open releasing the weights out the rear of the robot. The trapdoor will be made of MDF. This is shown in Figure 7.

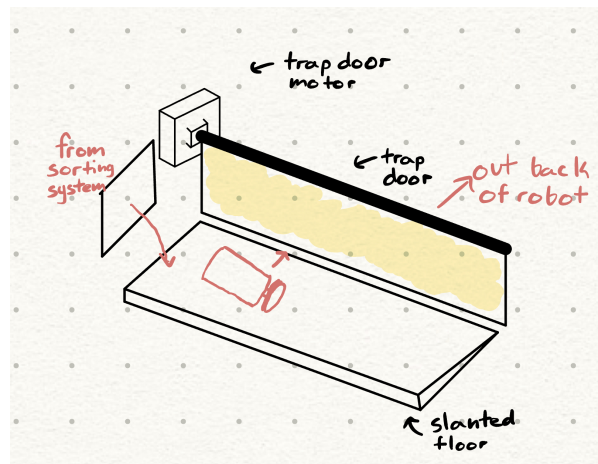


Figure 7: Storage system with trapdoor

4.1.5 Navigation

The robot will utilise two time of flight sensors to detect weights. These will be placed one above the other, facing out the front of the robot. The lower sensor will be set at a height which can detect weights. The higher sensor will be set at a height above the weights. When the lower time of flight sensor detects an object closer than the higher sensor, it will assume it is looking at a weight. Six IR sensors will surround the robot to detect objects. These will point outwards and sit above weights, so that weights are not confused with obstacles. The orientation of the sensors is shown in Figure ??.

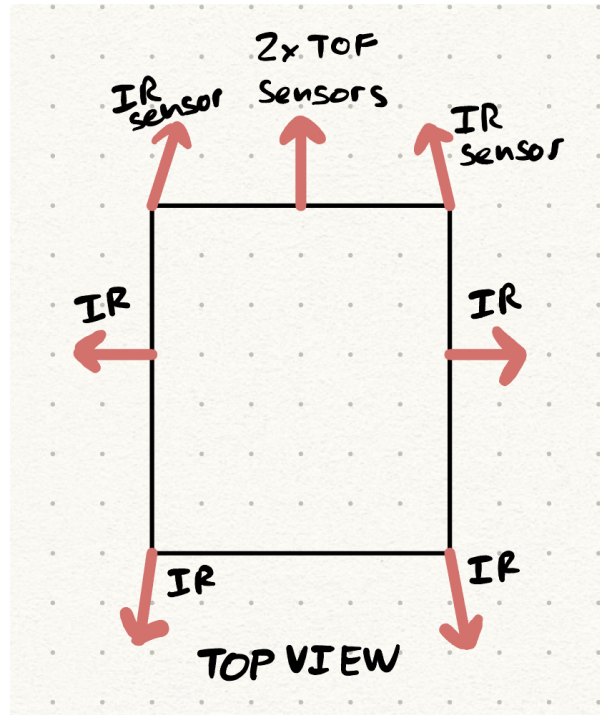


Figure 8: Orientation of sensors from top view

4.1.6 Movement

This design uses tank track on either side of the robot. A medium sized pulley has been attached to the motor to provide direct drive to the tracks. A large pulley wheel will be attached at the bottom front of the robot. This will allow the robot to clear obstacles including ramps and uneven surfaces, as it will have a larger approach angle. A small pulley will be attached to the furthest most back point of the robot, effectively eliminating departure angle. The use of tank tracks will minimise the risk of the robot becoming 'beached' due to breakover angle. This is shown in Figure 9.

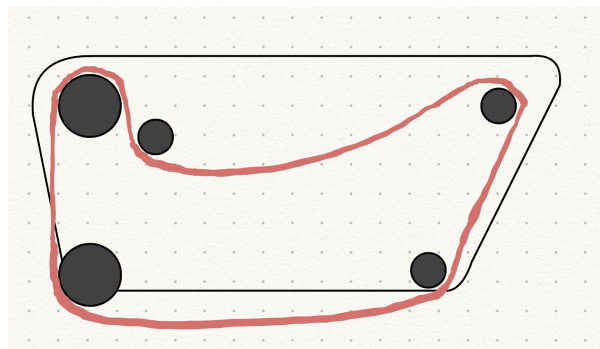


Figure 9: Tank track style movement system

4.2 Concept 2: Sucky Sucky 9000, Matthew Donnelly

4.2.1 General brief

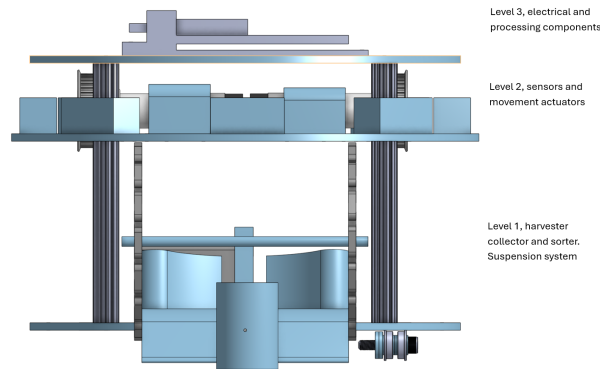


Figure 10: 3-level CAD design of Sucky Sucky 9000

This design was born from the necessity to remove the geometric constraints on the robot by creating an entirely new chassis design, which primarily involves a circular body making up the major design consideration in the concept. It would involve the use of the harvester mechanism, combined with a pinball-like system sorting based on weight and conductivity and using a 3-level layer structure as shown in Figure 10 above. The design is designed to be entirely modular to adapt to new issues and situations for the competition.

4.2.2 Weight collection and management

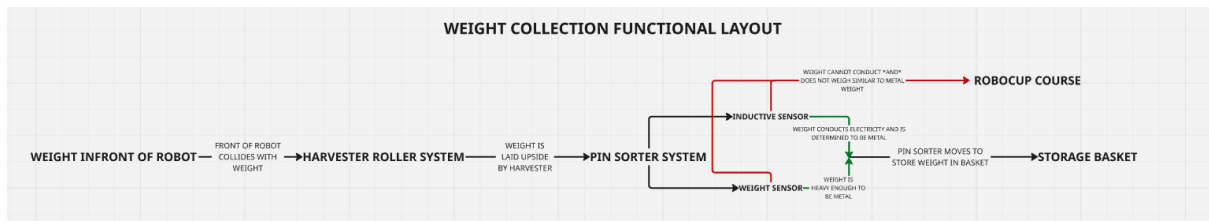


Figure 11: functional layout design of the weight capture and processing system for the robot

The current approach for the weight system, described in Figure 11, is when a weight is in front of the robot, a harvester-like system will pick up the weight and then throw such up a ramp, which will then land into a pinball sorting system. Both systems rely on JGY370 motors to power it. The pinball system would shift the weight into a horizontal orientation to allow the paddles to move the weight around easily.

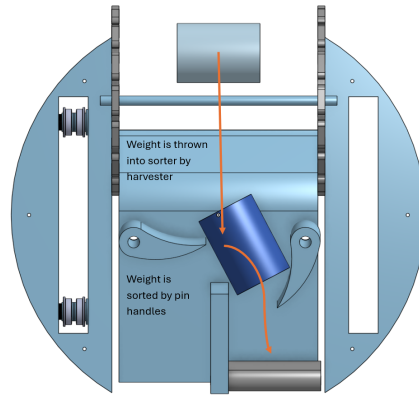


Figure 12: Pinball system shifting the weight from a rest position

The harvester design began from a simple ramp feature, with the wheel using rubber bands. However, from continual research and testing, the ramp design will become far more curved shape to be able to better control the weight's movement as shown in Figure 12. This system will be able to be used in case of bad orientation, which may reduce the effectiveness of one sensor type.

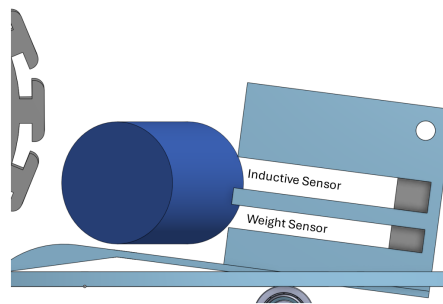


Figure 13: Inductive sensor as shown built into the divider wall

The use of a weight and inductive sensor would act as a 2-layer system to more accurately determine if a metal weight has been acquired, as shown in Figure 13. Once sorted, these weights will be ejected from the robot immediately or stored in a storage compartment that orients the weights to roll down the slope. This compartment has a flap door that would open and release the weights when back at home.

4.2.3 General movement and chassis design

The chassis design was considered to be a circle, to make it easier for it to escape corners, and be able to move around the collision poles and opponent robots, as observed in various rounds of the competition.

The robots' movement system would be actuated by a tank-track system, powered by FIT0277 motors, shown in Figure 14b, which would be housed inside the robot's chassis, shown in Figure 14a below. This maintains the circular shape for that geometric advantage, as well as reduce the chance of the track being thrown off by external factors. The design also allows for adjustable height for the track's tension. It was found in research that many groups had convoluted methods to make the older tracks work properly. This method keeps the design simple and provides that feature.

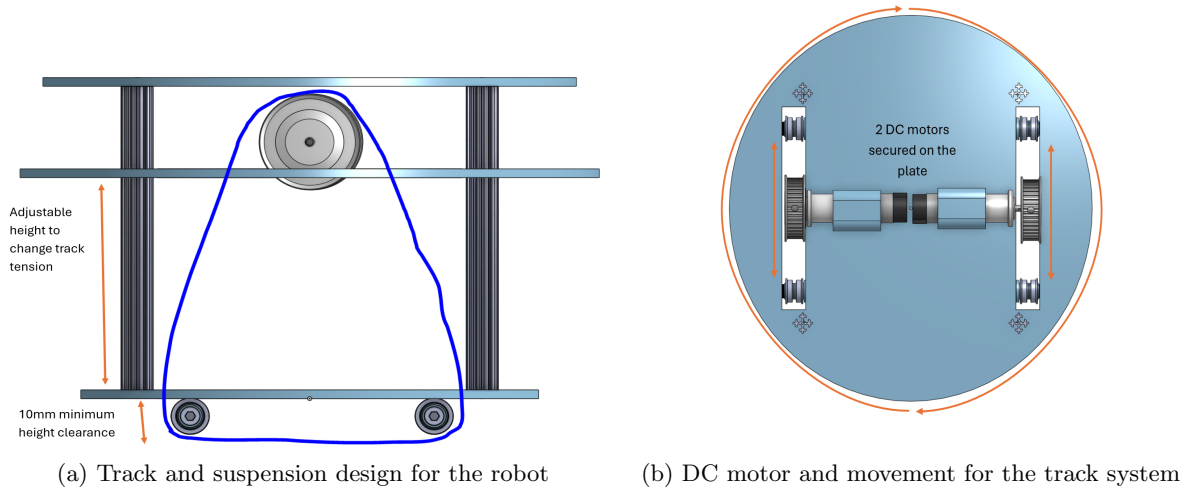


Figure 14: Description of the motor control design for the robot

4.2.4 External sensing and navigation

The approach to navigation in this initial design will rely on a Time Of Flight (TOF) and IR sensor combination. This would utilize the time of flight to be used as a front directional system on the lower sections of the design, being shown in Figure 15. Two sensors would need to be placed around the harvester, on the first level, to directionally work out where the weight is.

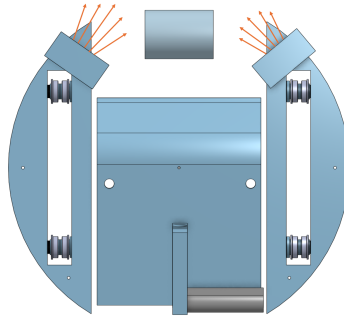


Figure 15: One Time Of Flight layout set for weight detection

The IR sensors would be laid out across the robot's circumference, and with the IR's constant range, be able to detect walls and other obstacles around it, acting as a compass of sorts to determine where obstructions are, as shown in Figure 16. This design would in turn be able to avoid such and still find the weights as it is offset far above the weight's height.

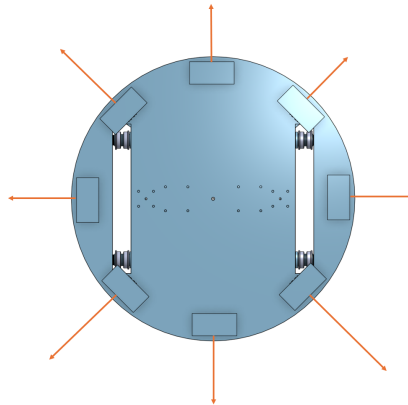


Figure 16: IR sensor layout on second level

4.3 Concept 3: Axiom Invictus, Dominic McNulty

4.3.1 General Brief

This design was created using information found in previous Robocup results reports. Findings from previous competitions prompted an extensive analysis of navigation methods, ensuring an efficient and robust system. The key features of this design includes a RFID scanner to sense NFC chips placed underneath the arena along with time-of-flight (TOF) sensors placed at a high and low point along the front of the robot to detect weights vs walls. A magnetic arm forms the basis of the collection system, depositing weights into a holding chamber with a floodgate-style release system when returning to home base. Navigation is handled using a taxicab geometry system, reducing navigation to movement between pre-placed RFID tags/nodes.

4.3.2 General movement

The design uses three of the provided Mecanum wheels for movement, configured equally spaced around a circle mounted on a triangle plate fabricated from MDF. The motivation for this approach was an initial test with tracks that presented challenges with varying tensions between each track side. The movement configurations in Figure 17 are a proposed solution to this problem.

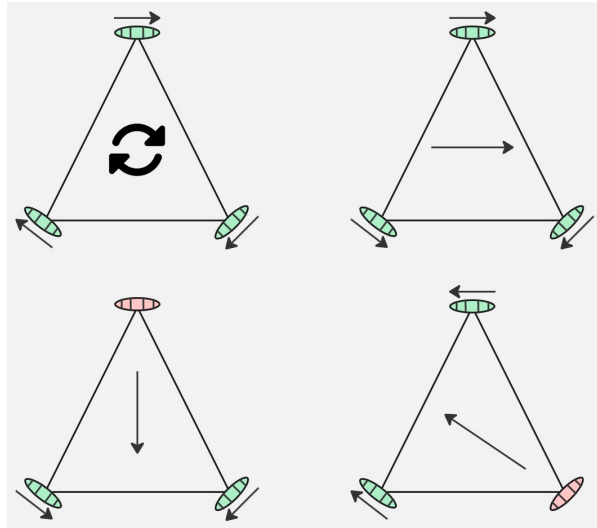
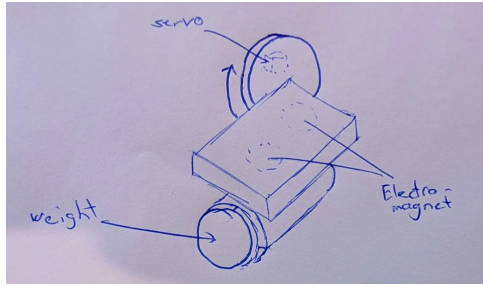


Figure 17: Movement configurations for the triangular Mecanum drive robot. **Top-left**: rotation; **Top-right**: rightward translation; **Bottom-left**: forward translation; **Bottom-right**: diagonal translation.

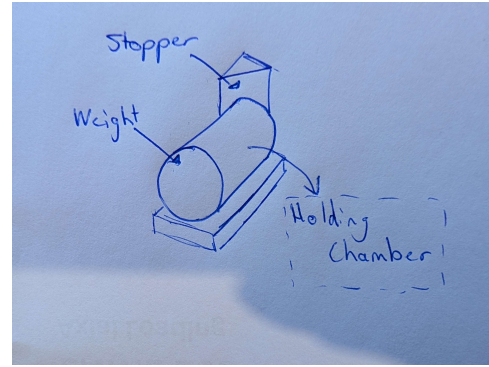
As presented in the figure above, during primary translational movement one wheel is un-powered, while the other 2 wheels are powered with rotation facing towards each other. This mechanism enables the robot to move in a straight line by balancing out the rotational forces acting on the chassis.

4.3.3 Weight Collection

Weight collection involves rotating a wheel and magnet system using the supplied high torque motor. The motor drives rotation in the wheel which causes a platform with two electromagnets to rotate 180° . The platform moves from a lower position, close to the ground, where the magnets face downwards to a higher position, where the magnets are facing upwards. Figure 18a shows the weight in the lower position with a weight attached. After rotation the platform is in an upward position; this movement forces the weight to roll into the holding chamber as shown in Figure 18b



(a) Weight before rotation



(b) Weight after rotation

Figure 18: Weight Collection System Demo

4.3.4 Weight Removal System

The weight removal system involves a floodgate-style flap that opens using a lead screw controlled by a motor. This allows all the weights to drop out of the holding chamber at once, making it a fast and efficient time. This was chosen as multiple returns to home base need to be made then the mechanism should be fast.

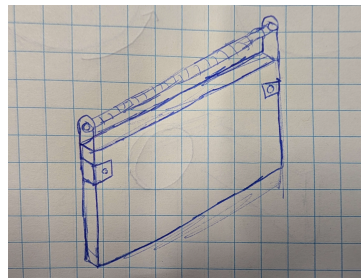


Figure 19: Floodgate Design

4.3.5 External sensing and navigation

Navigation uses a grid-based system that follows the RFID tags placed inside of the arena. The robot is designed to move between RFID tags with a sensor on the bottom of the robot, updating it's position each time a new RFID tag is read. This method is time independent and therefore does not succumb to accumulative error. Figure 20 shows the difference between using the manhattan distance vs euclidean distance measurement.

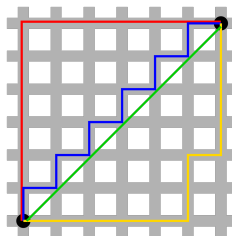


Figure 20: Manhattan Distance vs Euclidean Distance

The green line shows the usual distance measurement known as euclidean distance. The red, blue, and yellow show how the manhattan distance is a discretized version of euclidean distance, each line has the same overall length. This simplification will lead to a simplified code making it easier to understand the relative location of the robot at all times.

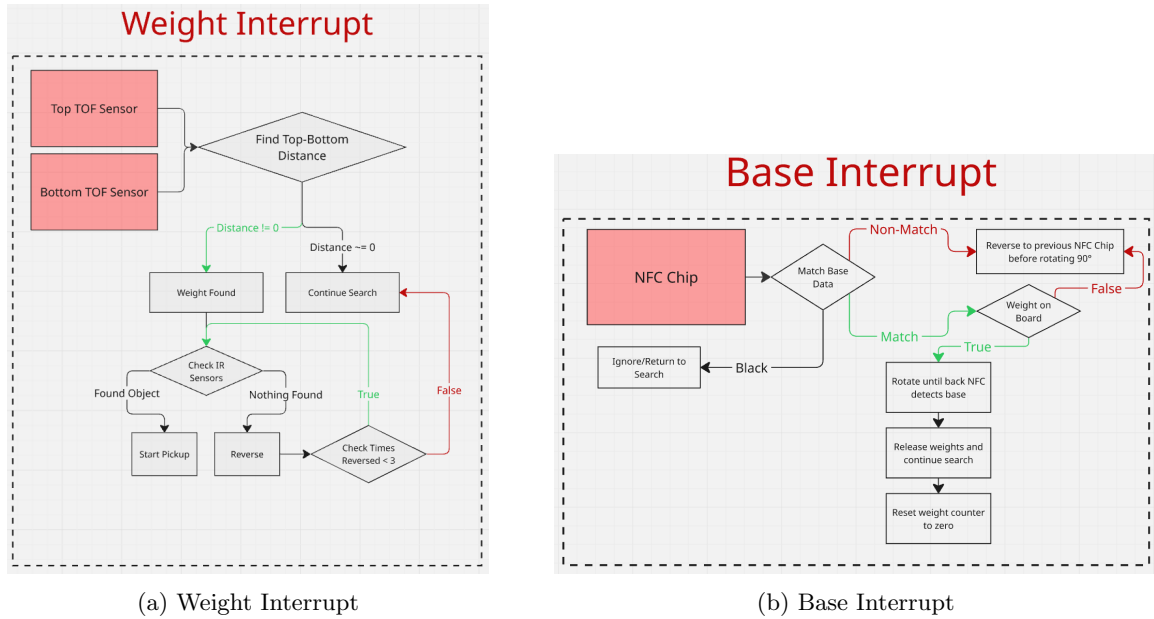


Figure 21: Two Interrupt Systems

Figure 21 shows the two types of interrupts that would be seen within code. The weight interrupt figure 21a shows how the two TOF sensors would be used to differentiate between walls and weights using the difference between the top and bottom sensor. The base interrupt figure 21b shows how the NFC chip will be sensed within the base leading to a variation in actions based on how many weights are currently detected to be within the holding chamber.

5 Preliminary Results

5.1 Harvester Test Bed experiment

For concept 2 specifically, the harvester weight collection system was created with an MDF cutout section to test its ability to move weights as shown in Figure 22. Furthermore, it was the perfect opportunity to try new approaches to the concept and also create a feasible approach that can be built around by the rest of the robot's design.



Figure 22: Harvester test bed constructed with the sample weight in front of the system

Initial tests were done on the placement and size of the harvester relative to the weight. Larger wheels inherently gave greater force onto the weight, but that force had to be initially concentrated in the groove of the weight to tip it over.

There was also some testing done on the ramp shape itself and its orientation to refine what should be used on the final design. Starting from a simple ramp angled at 30 degrees, the heavier weights could not go up the ramp without jamming or high effort from the motor. From research, various other groups used a 3d-printed gradient ramp, using smooth curved transitions to help the weight. The weight picking ability from all known orientations was 5 out of the 9 possible weight configurations. This would reduce the effectiveness of the harvester design unless significant testing and refinement was undertaken, which in this case may make the concept a time sink as compared to other design options.

5.2 General sensor testing and experimentation

The TOF, IR and Ultrasound sensors were tested in a rudimentary rig to understand the accuracy and effectiveness of the sensors if they were used for a robot's sensor apparatus, with results shown in Table 1.

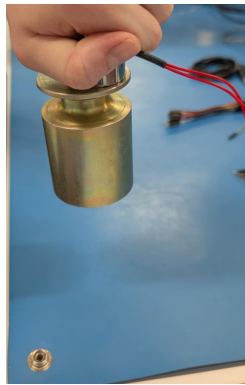
Table 1: Distances calculated by various sensors with relatively high accuracy from TOF and Ultrasound

Distances (mm)	IR sensor	TOF sensor	Ultrasound sensor
100	170	101	100
200	300	203	210
400	230	404	410
600	320	602	610
800	300	807	810

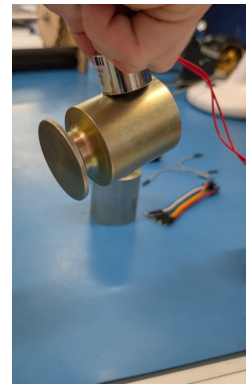
Testing of the IR sensor showed that it could not differentiate accurate object locations with the relative output data from the input distances of a weight. This would prove detrimental to using the IR in any way as an accurate form for detecting the weight. The TOF performed relatively accurately in determining the distance from the object, with a consistent 5 percent error in the distance calculations for the sensor. However, especially for finding a weight, such an inaccuracy at this current time is deemed unimportant for the robot's operation. The ultrasound was found through testing to be another accurate system. The ultrasound records data in 10mm chunks, with the accuracy only straying by 10mm at 800mm from the sensor. This resolution, however, would not have a major effect on the robot's necessary performance.

5.3 Electromagnet Testing

In terms of concept 3, the electromagnet was tested using the two main orientations that the design would be required to use when picking up the weight. Both orientations were tested using the heaviest weight to ensure the magnet would work on any kind of weight seen within the competition, this was done using the provided batteries running at 11V. The two orientations were vertically (original positioning) and horizontally (lying flat). These can be seen in Figure 23a and Figure 23b.



(a) Weight held vertically



(b) Weight held horizontally

Figure 23: Testing of Electromagnet Strength based on weight orientation

Results from the test showed that one electromagnet was sufficient in picking up both orientations of the weight vertically in a consistent manner. However, the horizontal weight was prone to knocks and bumps as applying any transverse force would usually result in the weight becoming detached.

5.4 Claw Collection System Test

To test the claw collection system of concept 1, and determine the variety of weight angle it could collect, a 3D printed concept claw was printed. Initial tests were done by placing weights in front of the claw in various orientations, and manually moving the claw to determine if it would pick the weight, or slip past. Various grip materials were tested on the claw to affect the friction between the claw and the weight. These results are displayed in the table below. Two weight orientations were able to be picked up using no grip and heatshrink as grip. While there was no quantitative difference shown, the heatshrink slipped less on the weights than no grip, as seen in Table 2.

Table 2: Table showing success of claw collection using different grips on heavy weight

Grip	Vertical	Flat (Longways)	Flat Sideways
No grip	Pass	Pass	Fail
Masking Tape	Pass	Fail	Fail
Heatshrink	Pass	Pass	Fail

6 Concept Evaluation

6.1 Concept Evaluation Matrix

A Figure Of Merit (FOM) table has been used to evaluate the effectiveness of each robot design. The strategy for evaluating the 3 designs centered on making each evaluation parameter a percentage as per the formulas in Appendix A and general logic. This was to more make each parameter equally judge each design. This percentage would then be processed to an associated weight, which was justified using Appendix B, to try to automatically normalize all evaluated parameters to make a further fairer judgment. When metrics were unable to be quantified, appropriate surrogate requirements were used. All evaluation parameters were calculated in Appendix C.

Table 3: Evaluation matrix table for all 3 concept designs

Concept Evaluation Matrix							
Criteria		Design 1:		Design 2:		Design 3:	
	Method weight	Normalised	Weighted	Normalised	Weighted	Normalised	Weighted
System's weight pickup ability	10	77.8%	7.78	55.6%	5.56	100%	10
System's snitch pickup ability	6	0%	0	100%	6	0%	0
Differentiate plastic at pickup	3	0%	0	0%	0	50%	1.5
System's manufacture ability	6	45%	3.6	69.3%	4.16	68%	4.08
Buildability	5	40%	2	50%	2.5	0.33	1.65
Weight collection time	7	45%	3.15	90%	6.3	50%	3.5
Traversability	9	100%	9	100%	9	71.4%	6.43
Power Usage	6	27.2%	1.63	30.1%	3.01	5.55%	0.33
Summed scores	52	52.2%	27.16	70.2%	36.5	52.9%	27.5

6.1.1 Concept 1

Concept 1 takes a durable and simplistic approach to its movement and chassis build. This is reflected as it received large scores for movement and buildability. The sorting and storage system also performs well against metrics.

This concept received low scores in metrics relating to its collection system. This is combined with a high complexity for the design of the collection system. As the robots performance is dependent on its ability to pick up weights, the failure to collect the snitch, and pick up weights from a wide range of angles severely limits the performance of this concept. This may be an important consideration to adjust for; which is reflected in this concepts overall score. Furthermore the concept's buildability and manufacturability were lower when compared to the other designs because of the excessive amount of components needed for the design to properly function.

6.1.2 Concept 2

The design's circular shape, the general configuration for the motors and suspension allowed for a high level of traversability. The design in this case would be able to go anywhere across the course, which

would reduce other future inefficiencies when the design further develops. The large use of MDF and its associated faster and easier fabrication and lighter weight makes the robot far easier to iterate upon compared to the other designs.

However, this design currently has an issue with buildability. The large number of sensors would cause greater complexity to organise the wires. The unique design meant a large amount of screws would be used before other methods of construction could be developed for future improvements. The method of weight collection, the harvester design, has several improvements that need to be made. However, the fast collection system paired with its acceptable weight orientations found within the testing conducted made it an effective system. Further refinement of the harvesters' ramp and overall design could lead to more orientations to be collected.

6.1.3 Concept 3

Concept 3 is a complex design using three geared motors for movement along with three TOF sensors on each side of the triangular shape in order to have precise navigation. The design has a great ability to manage weights of all different orientations whilst being easy to manufacture, the use of the electromagnet allows for the differentiation of some of the plastic weights giving it points in this area. One key downfall of this design is the power draw, which is caused by the use of three TOF sensors (800mA each) along with three motors for each Mecanum wheel. Another shortcoming is the amount of time required to pick up each weight. Picking up the weights requires the robot to be at a stand still to avoid knocks and bumps, slowing the robots effectiveness in the field. The traversability of a triangular design is a worry as corners of the arena could lead to problems with turning, especially due to the exterior wheel system.

7 Recommendations and Conclusion

Based on the concept evaluation matrix our team decided to choose concept 2 as a design. One key feature seen in this design was the fast and effective collection system, allowing for quick pickup of weights. Although concept 2 was not able to pickup weights in all orientations at this current point in time, the team believed that there was a high likelihood of fixing this issue in the future. Another important feature was the circular design of concept 2, allowing for traversal of all obstacles seen within the arena. This feature would significantly decrease the likelihood of a fatal error during operation as the robot would be able to move and rotate at any location inside the arena. Buildability was also seen to be a strong point with regards to concept 2. The use of MDF for the chassis proved to be one of the simplest materials to manufacture allowing for quick changes during the competition if needed. The navigation system seen in concept 3 was decided to be paired with the chassis and sensors seen in concept 2 in order to provide the best solution that the team could devise.

In conclusion for future development it is recommended to work mainly with concept 2, however to take some attributes from concept 3. Combining the two designs main features, the efficient pickup system and the strong navigation control, would lead to an extremely competitive and effective robot.

8 Appendix A Equation Set

$$Pickup\ Ability = \frac{unique\ orientations\ handled}{unique\ orientations} * 100 \quad (1)$$

$$Manufacturability = 1 - \frac{3DprintNum}{30} - \frac{MDFNum}{50} - \frac{SteelWaterJetNum}{10} \quad (2)$$

$$Traversability = \frac{unique\ obstacles\ handled}{Unique\ Obstacles} \quad (3)$$

$$Buildabilty = \frac{1}{0.1 * ScrewNum} + \frac{1}{0.5 * WireNum} \quad (4)$$

$$Weight\ Collection\ Time = 1 - CollectTime \times 0.05 \quad (5)$$

$$PowerUsage\ For\ Robot = \frac{4000 - Coeffic * SensorNum - Coeffic2 * MotorNum}{4000} \quad (6)$$

9 Appendix B Weight Justification

Evaluation Parameter	Weight	Justification
System's weight pickup ability	10	The robot needs to be able to encounter a weight and be able to pick it up no matter the orientation .
System's snitch pickup ability	6	The snitch is valuable, but can sacrifice performance for getting it.
Differentiate plastic at pickup	3	An important feature, however most robots have a plastic sorter feature internally.
System's manufacture ability	6	Different materials take longer times to make or require external help. Reducing these dependancies is important.
Buildability	5	There is a limited amount of time between rounds and so any failures or fixes that need to be implemented should be easy and quick
Weight collection time	7	Two minutes per round is a limited amount of time, if time needs to be spent specifically on picking a weight up it must be quick in order to continue searching for base or other weights
Traversability	9	The robot should be able to handle all forms of terrain encountered during a round, the arena is randomized between rounds with 7 unique defining obstacles.
Power Usage	6	This gives a baseline idea of the strain on the system for select components and their effects. These impacts may effect overall robot performance when under strain.

10 Appendix C Evaluation Matrix calculations

10.1 Pick Up calculations

From Experimentation: Concept 1 picked up 8/9 orientations, Concept 2 picked up 5/9 orientations, Concept 3 picked up 9/9 orientations.

10.2 Snitch pickup

From Experimentation: Only concept 2 can pick up the snitch, as the other concepts are not able to capture the snitch.

10.3 Plastic Weight differentiation

Only concept 3 could partially differentiate the plastic weight, as it depends on the orientation and if a weight with a metal top was encountered. None of the others had such abilities.

10.4 Manufacturability calculations

Concept 1: 6 3D printed components, 9 MDF components

$$1 - \frac{6}{30} - \frac{9}{50} = 0.66 \quad (7)$$

Concept 2 - 5 3D printed components, 7 MDF components

$$1 - \frac{5}{30} - \frac{7}{50} = 0.693 \quad (8)$$

Concept 3: 6 3D printed components, 6 MDF components

$$1 - \frac{6}{30} - \frac{8}{50} = 0.68 \quad (9)$$

10.5 Traversability calculations

Concept 1 - 7 out of 7 areas on the arena could be theoretically traversed.

Concept 2 - 7 out of 7 areas on the arena could be theoretically traversed by the design through testing.

Concept 3 - 5 out of 7 areas on the arena could be theoretically traversed.

10.6 Buildability calculations

Concept 1 -

$$\frac{1}{0.1 * 35} + \frac{1}{0.5 * 17} = 0.4 \quad (10)$$

Concept 2 - from arbitrary CAD and design consideration, there would be approximately 30 bolts and 12 sets of wires available, which then results in the following calculation:

$$\frac{1}{0.1 * 30} + \frac{1}{0.5 * 12} = 0.5 \quad (11)$$

Concept 3 - Approximately 52 bolts and 15 wires.

$$\frac{1}{0.1 \times 52} + \frac{1}{0.5 \times 15} = 0.33 \quad (12)$$

10.7 Weight Time Calculation calculations

Concept 1 - The weight can be lifted in 11 seconds.

$$1 - 11 * 0.05 = 0.45 \quad (13)$$

Concept 2 - Tested design found that the weight can be moved in less than 5 seconds then the robot is able to freely move from the test bed.

$$1 - 2 * 0.05 = 0.9 \quad (14)$$

Concept 3 - Weight was estimated to be picked up in 10 seconds without being able to move

$$1 - 10 \times 0.05 = 0.5 \quad (15)$$

10.8 Power Usage For Robot

This assumes that the robot will run for only 1 hour so we don't have to consider the hours the sensors will be used for:

All components used

TOF Sensors - 800mA

IR sensors - 50mA

Motors - 230mA

Geared Motors - 180mA

Magnet - 278mA when used at 11 volts (worst case)

Concept 1 - 2 TOF sensors (2*800mA), 6 IR sensors, 1 weight (inductive proximity), 2 geared motor (2*230mA), 3 geared motors (3*180mA)

$$PowerUsage For Robot = \frac{4000 - 1600 - 300 - 13 - 460 - 540}{4000} = 1087/4000 = 0.272 \quad (16)$$

Concept 2 - There are 2 TOF sensors (2*800mA), 8 IR sensors (Total of 400mA), 1 Weight Sensor (1.6mA), 1 Inductive Sensor (13mA) and 2 geared motors (2*230mA) and 2 geared Motors (2*180mA)

$$PowerUsage For Robot = \frac{4000 - 400 - 1600 - 1.6 - 13 - 460 - 320}{4000} = 2005.4/4000 = 0.501 \quad (17)$$

Concept 3 - There are 3 TOF sensors (3*800mA), 2 IR sensors (2*50mA), 3 geared motors (2*230mA), 2 geared motors (2*180mA) and the magnet (278mA)

$$Power Usage For Robot = \frac{4000 - 2400 - 100 - 460 - 540 - 278}{4000} = \frac{222}{4000} = 0.0555 \quad (18)$$